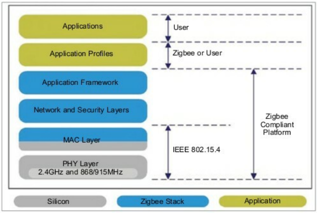

Fig. 2: ZigBee architecture (Image source: https://m.eet.com)

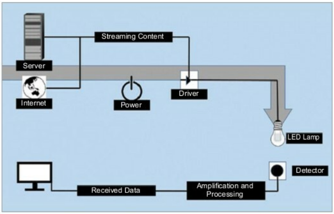

Fig. 3: Li-Fi architecture (Image source: www.focusearth.com)

RF-based communication technology. Mostly used for industrial applications, it has been assigned standard 802.15.4 by IEEE. ZigBee works in the same bandwidth as Wi-Fi, that is 2.4GHz, but consumes less power and offers a high level of security through 128-bit encryption. However, it enables limited data exchange.

ZigBee-based sensor networks are widely used in multilevel parking monitoring systems, warehouse measurement and control systems, and various environmental monitoring systems. Sensors are spread over the area to transmit information, and the products required to communicate are tagged in the range for flawless data travel.

IoT communication through light

Various companies and products use light as a medium to transfer data between various IoT objects. Li-Fi, IR and laser are three of the most popular IoT communication technologies in this category.

Li-Fi. Similar to Wi-Fi, Li-Fi uses optical wireless communication technology. It utilises visible, infrared and ultrasonic spectrum to transfer data. Li-Fi finds applications in underwater remotely-operated vehicles (as light is a better medium to communicate under water than radio frequencies)